



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

ID1063: Introduction to Programming (Lab-4)

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Lab-4 Questions

Four Questions

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Q1: The Alchemist-s Challenge

In a small, dusty workshop filled with the scent of strange herbs and bubbling liquids, a young apprentice named Harsh was learning the ancient art of alchemy. His master Saurabh, a wise alchemist, believed that the principles of logic were as important as magic itself.

One afternoon, the master presented Harsh with a challenge. He pointed to a rack with several rows of potion bottles.

"Harsh," the master began, "your task is to create a special glowing elixir. The process is precise."

He laid out the rules:

1. You will decide how many rows of bottles (N) to use for the elixir.
2. The first row has 1 bottle, the second has 2, the third has 3, and so on, up to the N^{th} row.
3. For each row, you must mix a specific amount of liquid from each bottle. The amount is equal to the bottle's position in that row.

For example, in Row 3, you'll take 1 ml from bottle 1, 2 ml from bottle 2, and 3 ml from bottle 3. The initial total for this row would be $1+2+3=6$ ml.

Here's the tricky part. After calculating the initial total of a row, you must check it against my Magic Number (M). If the row's total volume is less than M , you must add a special catalyst, 1 ml at a time, until the volume is less than M .

Q1: The Alchemist-s Challenge

"Before you waste any precious ingredients," the master said with a smile, "write a 'calculation scroll'—a C program—to figure out the final volume for each row. Let's see your logic at work!"

Harsh nodded, eager to prove his skills.

Input Format: Takes input as the total number of rows and the Magic Number, i.e., two integers separated by a space.

Output Format: Print "Final Volume for Row i is: j ml", where i is the row index and j is the final volume, both integers. You need to print N lines, one corresponding to each row, and every row must have a newline character at the end.

Example:

INPUT:

3 4

OUTPUT:

Final volume for Row 1 is: 4 ml

Final volume for Row 2 is: 4 ml

Final volume for Row 3 is: 6 ml

Q2: The Resonant Warding Spell

Years have passed. Harsh is no longer a mere apprentice but a skilled practitioner of logical alchemy. His master, seeing his potential, entrusts him with a task of immense importance: forging the final component for a grand warding spell.

"The warding requires a 'Prime Resonator'," the master explained, his voice low and serious. "Its power is determined by a very specific number. This number is the final echo in a chain of resonance starting from a Source."

He handed Harsh a crystal that pulsed with a faint light. "This crystal holds the Source (P). Your task is to find the Nth Resonant Echo derived from it."

The master laid out the arcane rules:

1. Start with the given Source, P.
2. Your goal is to find a sequence of N special prime numbers. Let's call the first one `prime_1`, the second `prime_2`, and so on.
 - `prime_1` is the first prime number found after P whose sum of digits is also a prime number.
 - `prime_2` is the second prime number found from P (after `prime_1`) whose sum of digits is also a prime number.
 - This pattern continues.
3. The Resonant Echo is the final result—the Nth special prime in this sequence.

Q2: The Resonant Warding Spell

"This is no simple task, Harsh," the master warned. "You will need to check if a number is prime, then sum its digits, and then check if that sum is also prime. You must repeat this, number by number, until you have found the entire sequence of N special primes. Create the calculation scroll before you begin."

The C Programming Problem: Your task is to write a C program to find the Nth Resonant Echo. The logic is complex and will require careful nesting of loops.

Input Format: Takes two integers, the Source (P) and the Sequence Number (N), separated by space.

Output Format: Print "The Resonant Echo is: j", where j is the resonant echo, integer.

Example

Input:

100 5

Output:

The Resonant Echo is: 139

Q2: The Resonant Warding Spell

Explanation:

- ❑ 101 (1st prime after 100): $1+0+1=2$ sum of digits prime $\rightarrow$ prime_1
- ❑ 103 (2nd prime after 100): $1+0+3=4$ sum of digits not prime $\rightarrow$ skip.
- ❑ 107 (3rd prime after 100): $1+0+7=8$ not prime $\rightarrow$ skip.
- ❑ 109: $1+0+9=10$ not prime $\rightarrow$ skip.
- ❑ 113: $1+1+3=5$ prime $\rightarrow$ prime_2
- ❑ 127: $1+2+7=10$ not prime $\rightarrow$ skip.
- ❑ 131: $1+3+1=5$ prime $\rightarrow$ prime_3
- ❑ 137: $1+3+7=11$ prime $\rightarrow$ prime_4
- ❑ 139: $1+3+9=13$ prime $\rightarrow$ prime_5 $\rightarrow$ Resonant Echo as this is Nth (5th) special prime after source.

Q3: The Bus Number Game

Aarav and Maya are newly admitted B.Tech students at IIT Hyderabad. Every morning at 9 AM, they wait at the Miyapur bus stop to board the college bus. But today, the weather is gloomy. Heavy rains have flooded the roads, and the bus is delayed. To kill time and distract themselves from the boring wait, they invent a quirky number game called The Bus Number Game.

Here are the rules of their game:

1. They decide a number N , which represents how many rounds they will play.
2. In each round:
 - If N is even, they write 4 consecutive numbers but skip the first even number in that round.
 - If N is odd, they write 3 consecutive numbers but skip the first odd number in that round.
3. At the end of each round, they write the word "bus".

Your task is to simulate their game and print exactly what Aarav and Maya would write on the wall while waiting for the bus.

Input format: Only one line containing a number N ($1 \leq N \leq 1000$)

Output format: Print N lines according to the rules above. Each line must end with the word bus followed by a newline character. There is a single space in between every number and the word "bus".

Q3: The Bus Number Game

Example 1:

Input:

4

Output:

1 3 4 5 bus

7 8 9 10 bus

11 13 14 15 bus

17 18 19 20 bus

Example 2:

Input:

5

Output:

2 3 4 bus

6 7 8 bus

10 11 12 bus

14 15 16 bus

18 19 20 bus

Q4: Earth is under attack

The year is 3050. Earth is under attack from two cosmic warlords—Draconia and Hydratos, the long-lost siblings of Lord Zedd. Each commands an enormous fleet of planet-ships. The Power Rangers have been summoned for one final mission to save Earth from total annihilation.

Draconia's fleet consists of D planet-ships, while Hydratos's fleet consists of H planet-ships. Both D and H are very, very large integers—so large that ordinary strategies won't work. Fortunately, Zordon equips the Rangers with two powerful weapons:

1. The **Twin Blaster** destroys exactly one planet-ship from both fleets simultaneously (only possible if both fleets still have at least one planet-ship).
2. The **Amplifier Ray** focuses on one fleet of your choice, splitting each of its ships into two, thus doubling the number of planet-ships in that fleet.

The Rangers must figure out a way to destroy both fleets completely. If it is possible, you must calculate the minimum number of weapon firings required. If not, output -1.

Given two very large integers D and H (the number of planet-ships in Draconia's and Hydratos's fleets), determine whether it is possible for the Power Rangers to destroy both fleets. If yes, print the minimum number of weapon firings required. Otherwise, print -1.

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Given two very large integers D and H (the number of planet-ships in Draconia's and Hydratos's fleets), determine whether it is possible for the Power Rangers to destroy both fleets. If yes, print the minimum number of weapon firings required. Otherwise, print -1.

Input Format: The input consists of a single line containing two very large integers D and H separated by a single space.

Output Format: Print a single large integer: the minimum number of gun firings required to destroy both fleets, or -1 if it is impossible. There is a newline character at the end of the line.

Q4: Earth is under attack

Constraint: $0 \leq D, H \leq 10000000000000$

Example 1:

Input:

10 10

Output:

10

Example 2:

Input:

0 5

Output:

-1

Example 3:

Input:

2 9

Output:

12

Example 4:

Input:

13 14

Output:

15

Note: Use **long long int** for the operations.